

SARA SHARMA

Data Scientist | Advanced Analytics | AI/ML & Data Products

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SUMMARY

Data Scientist and AI/ML practitioner experienced in advanced analytics, predictive modeling, statistical analysis, and decision support across IIT Bombay, DRDO, NITI Aayog, and industry. Skilled in Python, SQL, forecasting, model evaluation, RAG, prompt engineering, and AI/ML orchestration. Translates business and operational problems into actionable analytical solutions and measurable outcomes while collaborating across technical workstreams, with experience in model reliability, explainability, and operationalization.

EDUCATION

SRM Institute of Science and Technology

Chennai, India

Bachelor of Technology (B.Tech) in Computer Science & Engineering (AI & ML)

2026

CGPA: 8.18/10

EXPERIENCE

Defence Research and Development Organisation (DRDO)

Hyderabad, India

AI/ML Research Volunteer

2026 – Present

- Conduct AI/ML research under scientist supervision, integrating heterogeneous data sources and developing analytical solutions for object and event recognition.
- Improve model specificity through iterative experimentation, performance evaluation, and analysis across varied operating conditions.
- Achieved **90% accuracy in initial testing** and continue refining model performance, object/event recognition, and evaluation across varied operating conditions.

Indian Institute of Technology Bombay

Mumbai, India

AI/ML Intern

Sept 2026 – Present

- Curated and augmented **100+ datasets in two weeks** for spatial scene understanding; developed, trained, and evaluated CNN and Swin Transformer models using accuracy and class-wise performance metrics.
- Built **data-quality, synthetic-degradation, and loss-scoring pipelines**, conducting deep-dive analysis, error analysis, model evaluation, and quantitative assessment to identify performance issues and improve analytical reliability.

NITI Aayog, Government of India

New Delhi, India

Data Management & Analysis Intern

Aug 2026 – Sept 2026

- Analyzed and integrated **16 government datasets covering 900K+ records** to develop an AI-enabled decision-intelligence platform for beneficiary profiling, predictive scoring, intervention recommendation, and outcome monitoring.
- Developed an end-to-end analytical workflow spanning risk/success scoring, blocker identification, program matching, outcome projection, case prioritization, and action-plan generation, producing **3–5 success factors, 2–4 blockers, and 2–5 targeted interventions per profile**.
- Conducted deep-dive analysis of PMMY, PLFS, and Udyam data across credit, employment, and enterprise activity; applied Logistic Regression and Random Forest to generate actionable insights and support data-driven decision-making.

Defence Research and Development Organisation (DRDO)

Hyderabad, India

AI Research Intern

Jan 2026 – Jul 2026

- Designed and prototyped **ORACLE-X**, an uncertainty-aware AI/ML solution integrating temporal Transformers, state estimation, physics-guided modeling, and XGBoost for predictive anomaly detection.
- Developed **end-to-end ML pipelines** spanning telemetry preprocessing, feature engineering, anomaly scoring, uncertainty estimation, explainability, and fault analysis; evaluated robustness across operating regimes and sensor-degradation scenarios.

Fluentgrid Limited

Visakhapatnam, India

Python & Machine Learning Intern

Jun 2025 – Sept 2025

- Developed a **5-engine XGBoost forecasting system** integrating demand, risk, and scenario analysis; engineered predictive features and applied SHAP for model explainability and analytical decision support.
- Analyzed operational datasets to identify business patterns, performance drivers, demand behavior, and potential risks, translating findings into quantitative insights for operational decision-making.

Inferrix Greentech Pvt. Ltd.

Noida, India

AI Research Intern

Jul 2024 – Sept 2024

- Developed a **Physics-Informed Neural Network** integrating HVAC, weather, building, and IoT data for predictive energy modeling and data-driven decision support.
- Generated heat-load profiles across operating conditions, contributing to a reported **15% reduction in energy consumption** through predictive modeling and analytical optimization.

SELECTED PROJECTS

E-Commerce Campaign Performance Prediction & Optimization | Python · Scikit-learn · React

- Built a **prescriptive analytics solution** achieving **82% prediction accuracy** for customer-volume and sales-surge detection, translating predictive models into actionable business insights.
- Automated **65% of manual analysis** through optimization logic for campaign budgets, targets, and timing; developed an analytics interface for KPI monitoring and data-driven decision support.

ML Model Reliability & Risk Under Distribution Shift | Model Evaluation · LLM Testing

- Developed an **end-to-end ML reliability framework** using uncertainty estimation, calibration, and out-of-distribution detection to evaluate model performance under temporal and spatial distribution shifts.
- Compared standard, uncertainty-aware, and abstention-based models through quantitative evaluation, extending testing to **LLM-based systems under adversarial and OOD inputs**.

RESEARCH & PUBLICATION EXPERIENCE

ECOAVIA: Aviation Demand Forecasting | Presented/Publication pending.

ICISC 2026

- Developed a hybrid **XGBoost-ARIMA-LSTM forecasting framework**, integrating climate and infrastructure variables to analyze demand trends and generate data-driven insights for sustainability-oriented policy analysis.

SKILLS

Programming & Analytics: Python, SQL, PostgreSQL, Pandas, NumPy, EDA, Advanced Analytics, Statistical Analysis

Machine Learning & AI: Predictive Modeling, Regression, XGBoost, Scikit-learn, PyTorch, TensorFlow, Deep Learning, Multimodal AI

AI/LLM Solutions: LLM Applications, RAG, Prompt Engineering, LangChain, LangGraph, MCP, AI/ML Orchestration

Modeling & Evaluation: Feature Engineering, Model Evaluation, Error Analysis, Model Reliability, Uncertainty Estimation, Anomaly Detection, Explainable AI, SHAP, Forecasting

Data Products & Engineering: ETL, Data Pipelines, FastAPI, Docker, MLflow, Git, CI/CD

Decision & Business Analytics: Decision Intelligence, Predictive/Scenario/KPI/Strategic Analytics, Data-Driven Decision-Making, Actionable Insights

Visualization: Power BI, Tableau, Matplotlib, React

LEADERSHIP & ACHIEVEMENTS

- Presented ECOAVIA, an AI-driven hybrid forecasting study for sustainable aviation, at **ICISC 2026**.
- Led **NITI Aayog research workstreams**, coordinating analytical tasks and consolidating findings across the team.
- Delegate, VIT Model United Nations 2024; part of the SRMIST team awarded **Best Delegate Team**.